

The kernel chosen by GRUB mounts the root file system defined in the `grub.conf` file during this step of the boot process. After that, it runs the `/sbin/init` program, which is always the first code to run. Its process id (PID), should always be 1. The kernel then uses Initial RAM Disk (`initrd`) to create a temporary root file system until the real file system is mounted.

```

[ OK ] Started Apply Kernel Variables.
[ OK ] Mounted Kernel Debug File System.
[ OK ] Mounted Huge Pages File System.
[ OK ] Mounted USBX Message Queue File System.
[ OK ] Started Read and set NFS filesystems from /etc/sysconfig/network.
[ OK ] Activated uapd /dev/wapper/c1-uapd.
[ OK ] Reached target Smp.
[ OK ] Started Network Boot and Kernel File Systems.
      Starting Flash Journal to Persistent Storage...
      Starting Load/Save Random Seed...
      Starting Create Static Device Nodes in /dev...
[ OK ] Started Load/Save Random Seed.
[ OK ] Started Flash Journal to Persistent Storage.
[ OK ] Started Setup Virtual Console.
[ OK ] Started Create Static Device Nodes in /dev.
      Starting udev Kernel Device Manager...
[ OK ] Started udev Kernel Device Manager.
[ OK ] Created slice system-usb/lun2/lun2gpuslice.slice.
      Starting LUN event activation on device 0:2.
[ OK ] Started Monitoring of LUN mirrors, snapshots etc. using dmccard or progress polling.
[ OK ] Reached target Local File Systems (Pre).
      Starting File System Check on /dev/disk/by-uuid/8068ca58-6212-484b-9148-b512c612c58a...
[ OK ] Started LUN event activation on device 0:02.
[ OK ] Started File System Check on /dev/disk/by-uuid/8068ca58-6212-484b-9148-b512c612c58a.
      Mounting /boot...
[ OK ] Mounted /boot.
[ OK ] Reached target Local File Systems.
      Starting Import network configuration from initramfs...
      Starting Tell Plymouth To Write Out Runtime Data...
      Starting Restore /run/initramfs on shutdown...
[ OK ] Started Restore /run/initramfs on shutdown.
[ OK ] Started Tell Plymouth To Write Out Runtime Data.
[ OK ] Started Import network configuration from initramfs.
      Starting Create Volatile Files and Directories...
[ OK ] Started Create Volatile Files and Directories.
      Starting Security Auditing Service...

```

Source: <https://www.tecmint.com/>

Kernel Booting Up

5. Init

Now, your system executes run-level programs. At some point, it will search for an `init` file. This is generally found at `/etc/inittab`, and it helps to decide the Linux run level.

6. Runlevel programs

Depending on whatever Linux distribution you've installed, you could notice several services starting up. You could, for example, catch yourself typing `Sendmail`. It's worth noting that the precise placement of these directories changes as per distribution. The programs begin with an 'S' or 'K' where 'S' stands for startup and 'K' stands for the kill, respectively, in the various run level folders. Thus, during system start, startup programs are performed, and kill programs are executed during the shutdown.